

Incident Investigation Report

Executive Summary

On 09 June 2026, a security alert was generated within the Splunk SIEM platform indicating abnormal authentication activity against an Ubuntu Linux server. Initial triage identified multiple failed SSH authentication attempts originating from a single external source IP address. The observed behavior was consistent with a brute-force password guessing attack targeting privileged user accounts.

The Security Operations Center (SOC) initiated an investigation to determine the scope, impact, and legitimacy of the activity. Analysis confirmed repeated failed authentication attempts without evidence of successful compromise. The attack was contained and documented for future detection engineering improvements.

Incident Metadata

Field	Value
Incident ID	SOC-2026-001
Incident Type	SSH Brute Force
Severity	Medium
Status	Closed
Detection Source	Splunk SIEM
Affected Asset	Ubuntu SOC Lab Server
MITRE ATT&CK Technique	T1110.001 – Password Guessing
Analyst	Mohammed Marvan
Investigation Date	09 June 2026

Detection Details

Alert Name

Multiple Failed SSH Authentication Attempts

Detection Logic

```
index=* "Failed password"
| stats count by src,user
| where count > 5
```

Alert Trigger

The alert triggered after more than five failed authentication attempts were observed from the same source IP against the same target account.

Investigation Methodology

The investigation followed the SOC incident response lifecycle:

- 1. Alert Validation
- 2. Event Triage
- 3. Scope Determination
- 4. Evidence Collection
- 5. Threat Assessment
- 6. Containment Verification
- 7. Documentation

Timeline of Events

Timestamp	Event
18:10 UTC	Initial failed authentication attempt detected
18:11 UTC	Multiple failed login attempts observed
18:12 UTC	SIEM alert generated
18:13 UTC	Analyst initiated investigation
18:18 UTC	Source IP identified
18:25 UTC	Activity classified as brute-force attempt
18:30 UTC	Investigation completed

Evidence Collection

Query 1 – Failed Authentication Events

```
index=* "Failed password"  
| table _time src user host
```

Findings

- Source IP repeatedly attempted authentication.
- Multiple username combinations were observed.
- Authentication attempts occurred within a short timeframe.

Query 2 – Successful Authentication Verification

```
index=* ("Accepted password" OR "Failed password")  
| table _time src user
```

Findings

No successful authentication events were identified from the attacking source IP.

Indicators of Compromise (IOCs)

IOC Type	Value
Source IP	192.168.1.100
Target Host	ubuntu-soc-lab
Protocol	SSH
Port	22
Activity	Password Guessing

MITRE ATT&CK Mapping

Tactic

Credential Access

Technique

T1110.001 – Password Guessing

Justification

The attacker repeatedly attempted authentication using multiple password guesses against a valid account through the SSH service.

Impact Assessment

Confidentiality

No impact observed.

Integrity

No unauthorized changes identified.

Availability

No disruption of service detected.

Overall Risk

Medium

The activity demonstrates active adversary behavior but did not result in system compromise.

Root Cause Analysis

The root cause of the incident was exposure of an SSH service accessible from an untrusted network combined with the absence of rate limiting controls.

No evidence suggests credential compromise or successful privilege escalation.

Containment Actions

- Monitored source IP activity.
 - Verified absence of successful authentication.
 - Recommended firewall restrictions for SSH access.
 - Recommended implementation of Fail2Ban.
 - Recommended multi-factor authentication for administrative accounts.
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Lessons Learned

1. SSH authentication failures should be continuously monitored.
 2. Brute-force detection thresholds should be tuned for rapid alerting.
 3. External SSH exposure should be minimized.
 4. MFA should be implemented for administrative access.
 5. Threat detection content should be mapped to MITRE ATT&CK.
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Analyst Conclusion

Investigation confirmed malicious brute-force behavior targeting SSH services on the Ubuntu server. The activity was successfully detected by Splunk SIEM and investigated according to SOC procedures. No successful compromise was identified. Existing controls were effective in preventing unauthorized access; however, additional hardening measures are recommended to reduce future attack exposure.

Final Classification: Confirmed Brute Force Attempt

Incident Status: Closed